Employee Burnout Detection

using Machine Learning & HR Analytics

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Introduction & Problem Statement

What is Employee Burnout?

- A state of chronic physical and emotional exhaustion
- Caused by prolonged workplace stress and overwork
- Recognized by WHO as an occupational phenomenon
- Affects productivity, mental health, and job satisfaction
- Often goes undetected until it causes serious damage

Why Early Detection Matters?

- 77% of professionals have experienced burnout (Deloitte)
- Leads to high attrition, absenteeism, and health costs
- Organizations lose billions annually due to burnout
- ML can detect patterns before burnout becomes critical
- Timely HR intervention can save careers and costs

Objectives & Dataset

Project Objectives

- Identify key factors driving employee burnout
- Visualize burnout trends across work patterns
- Cluster employees into risk groups using K-Means
- Predict burnout risk categories using Random Forest
- Provide actionable insights to HR departments

Dataset Overview

250 Employee Records

Key Variables:

- Working Hours per Week
- Overtime Hours per Week
- Task Load Score
- Leave Days in Last 6 Months
- Average Sleep Hours
- **Burnout Risk Category (Target)**

Project Methodology / Workflow

1

Data Loading & Preprocessing

Import dataset, handle missing values, encode categories, normalize features

2

Data Visualization

Plot burnout distribution, boxplots, heatmaps to uncover patterns

3

K-Means Clustering

Group employees into clusters based on work patterns (unsupervised)

4

Random Forest Classification

Train model to predict burnout risk category (supervised learning)

5

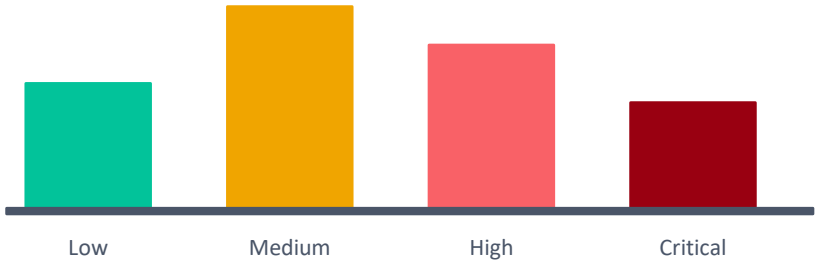
Model Evaluation

Confusion matrix, accuracy report, and feature importance analysis

Data Visualization Insights

Graph 1: Burnout Risk Distribution

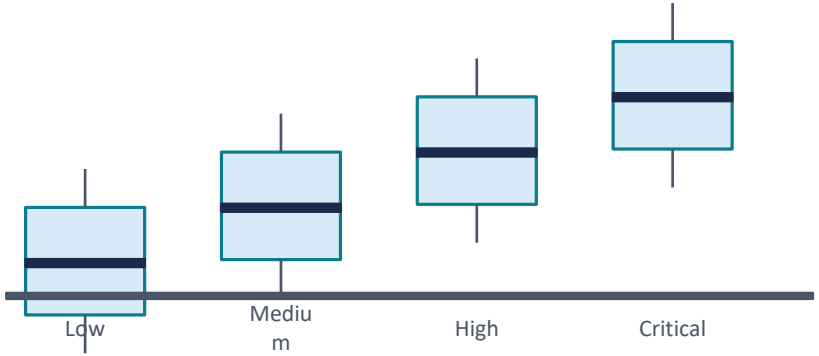
- Medium risk is the most prevalent category
- ~20% employees fall in High or Critical risk zone



Burnout Risk Distribution Bar Chart

Graph 2: Working Hours vs Burnout Risk

- Higher working hours strongly correlate with burnout
- Critical risk employees work 55+ hours per week



Working Hours vs Burnout Risk Boxplot

Clustering Analysis — K-Means

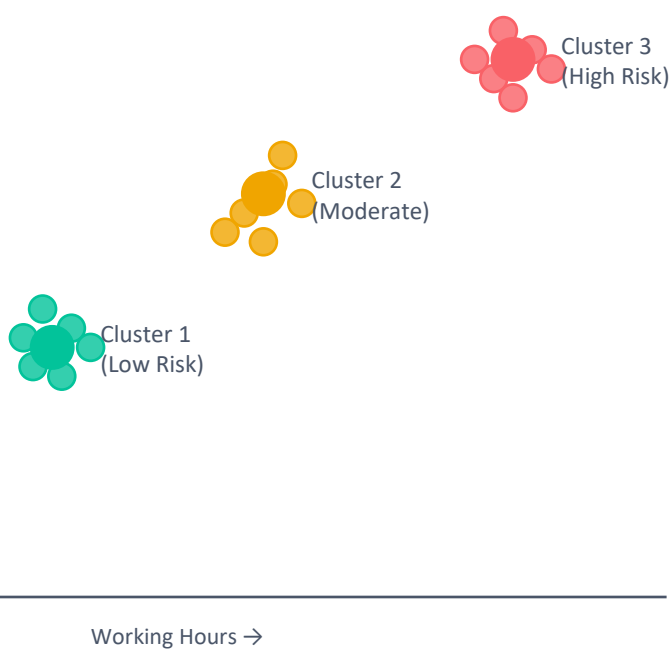
What is K-Means Clustering?

- An unsupervised machine learning algorithm
- Groups employees with similar work patterns into clusters
- No prior labels needed — purely data-driven grouping
- Used K=3 clusters based on Elbow Method analysis

Clustering Features Used:

- Working Hours per Week
- Overtime Hours per Week
- Task Load Score

Cluster Visualization



[Insert: K-Means Cluster Graph]

Random Forest Model & Results

How Random Forest Works

- An ensemble of multiple decision trees
- Each tree votes on the burnout risk category
- Final prediction = majority vote across all trees
- Handles class imbalance and noisy data well
- 80/20 train-test split used for training
- All 6 HR variables used as input features

Model Results

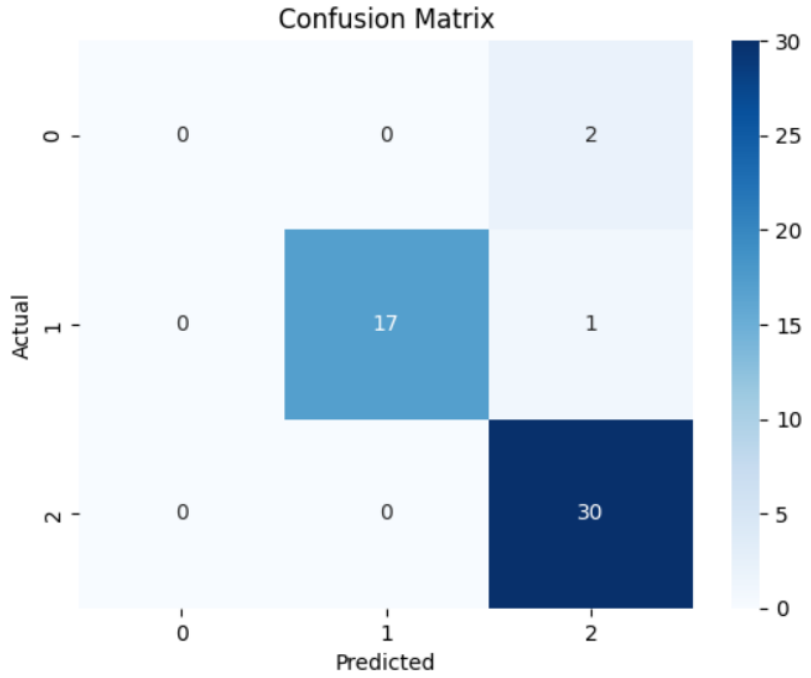
~94%

Model Accuracy

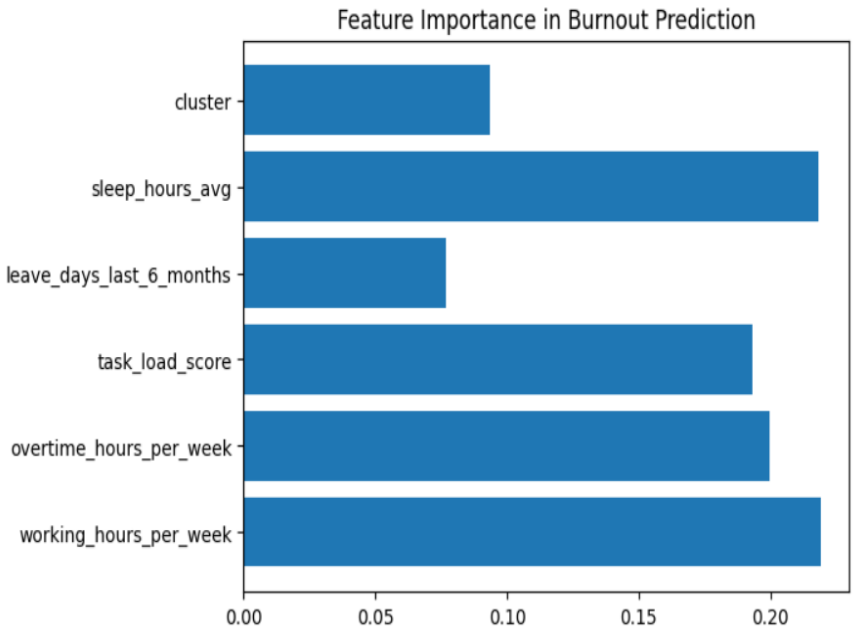
- Correctly classifies 94 out of 100 employees
- High precision for High & Critical risk categories
- Robust performance across all four risk categories
- Outperforms simpler models like Logistic Regression

Model Evaluation & Feature Importance

Confusion Matrix & Classification Report



Feature Importance



Impact & Conclusion

Early Detection

Identify at-risk employees before burnout becomes critical

Smarter HR Decisions

Data-driven insights replace guesswork in people management

Cost Reduction

Reduce attrition, absenteeism, and healthcare-related costs

Employee Wellbeing

Proactive support improves morale, retention, and productivity

This project demonstrates that ML and HR Analytics can work together to create smarter, more human workplaces.